

Review

Digital Twins Generated by Artificial Intelligence in Personalized Healthcare

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Featured Application: The potential application of the work relates to novel digital twins of patients or their organs based on artificial intelligence.

Abstract: Digital society strategies in healthcare include the rapid development of digital twins (DTs) for patients and human organs in medical research and the use of artificial intelligence (AI) in clinical practice to develop effective treatments in a cheaper, quicker, and more effective manner. This is facilitated by the availability of large historical datasets from previous clinical trials and other real-world data sources (e.g., patient biometrics collected from wearable devices). DTs can use AI models to create predictions of future health outcomes for an individual patient in the form of an AI-generated digital twin to support the rapid assessment of in silico intervention strategies. DTs are gaining the ability to update in real time in relation to their corresponding physical patients and connect to multiple diagnostic and therapeutic devices. Support for this form of personalized medicine is necessary due to the complex technological challenges, regulatory perspectives, and complex issues of security and trust in this approach. The challenge is also to combine different datasets and omics to quickly interpret large datasets in order to generate health and disease indicators and to improve sampling and longitudinal analysis. It is possible to improve patient care through various means (simulated clinical trials, disease prediction, the remote monitoring of a patient's condition, treatment progress, and adjustments to the treatment plan), especially in the environments of smart cities and smart territories and through the wider use of 6G, blockchain (and soon maybe quantum cryptography), and the Internet of Things (IoT), as well as through medical technologies, such as multiomics. From a practical point of view, this requires not only efficient validation but also seamless integration with the existing healthcare infrastructure.

Keywords: artificial intelligence; machine learning; medical engineering; personalized medicine; digital twin; DT; AI-based DT; omics



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1. Introduction

The concept of digital twins as virtual representations of physical objects or systems has a history dating back several decades [1,2]. Although the term “digital twin” only gained popularity 20 years ago, its basic concepts have been around for over 60 years. Key milestones in the history of digital twins include:

1. Early foundations (1960s–1980s);

2. Development of the digital twin concept (1990s–2000s);
3. Technological advances (2010s);
4. Modern applications (2020s) [2–4].

The concept of using computers to simulate physical systems began in the 1960s and 1970s with the development of computer-aided design (CAD) systems that allowed the creation of digital models of physical objects. The early concept of DTs is believed to be partially related to NASA’s use of pairing technology to create mirror systems during the Apollo missions, which created physical and digital replicas of spacecraft on Earth, enabling the monitoring and troubleshooting of space missions. In the 1990s, the development of product lifecycle management (PLM) systems integrated product lifecycle data and processes into digital models for design, manufacturing, and maintenance. The term “digital twin” was first used by Dr. Michael Grieves at the University of Michigan in 2002 as a virtual representation—a real-time digital equivalent of a physical object or process. The development of the Internet of Things in the 2010s extended the capabilities of DTs to include sensors embedded in physical objects. These enabled real-time data collection and communication with their digital counterparts. During this period, digital twins have been implemented in many industries, including healthcare, as discussed later. Subsequently, the integration of AI and machine learning (ML) with DTs has enabled more sophisticated analysis and predictive capabilities. As a result, DTs have evolved from static models to dynamic systems capable of simulating complex behaviours and scenarios. In 2020, the Digital Twin Consortium was established to promote best practices and standardize digital twin technologies across technology companies, government agencies, and academic institutions. The further development of AI-based DTs by 2030 will be based on emerging technologies, including those based on 6G (Figure 1) [4–6].

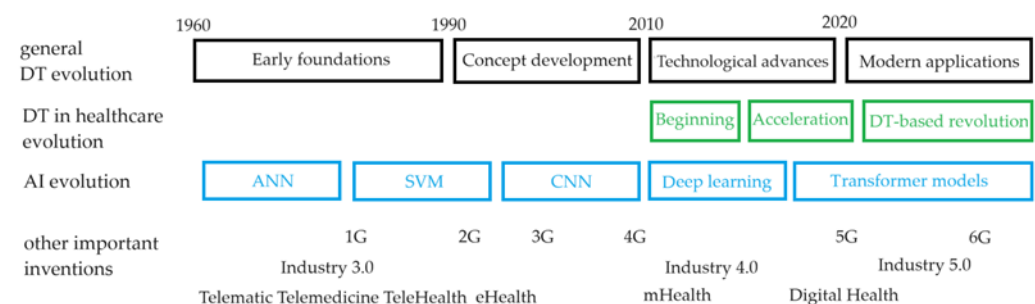


Figure 1. DTs development against the background of AI development (own version).

The story of digital twins in healthcare began to gather momentum in the 2010s, with the rise of the Internet of Things and artificial intelligence (AI)-based data analytics enabling the real-time monitoring of patient health through wearable devices. Early applications focused on creating digital replicas of organs, such as the heart, to simulate conditions and optimize treatment plans. The concept has been further developed through the integration of AI and machine learning, enabling personalized medicine through predictive modelling of patient outcomes. By the mid-2010s, digital twins were being used in surgical planning, where virtual models of a patient’s anatomy helped surgeons practice and refine procedures before actual surgeries. The COVID-19 pandemic has accelerated interest in digital twins as a tool to manage and predict the spread of the virus and assess the impact of public health interventions. Advances in genomics have also enabled the creation of digital twins at the cellular level, facilitating more targeted therapies. Hospitals began implementing digital twins of entire facilities to optimize patient flow and resource management, improving overall efficiency. This concept extended to drug development, where virtual models of human physiology were used to simulate drug interactions, reducing the time and cost of clinical trials. The increasing availability of high-quality medical data and advanced computing power have made digital twins more accessible and accurate, driving their widespread adoption. From the 2020s, AI-based DT technologies in healthcare are poised to

revolutionize patient care, offering unprecedented levels of personalization and predictive power (Figure 2) [7–11].

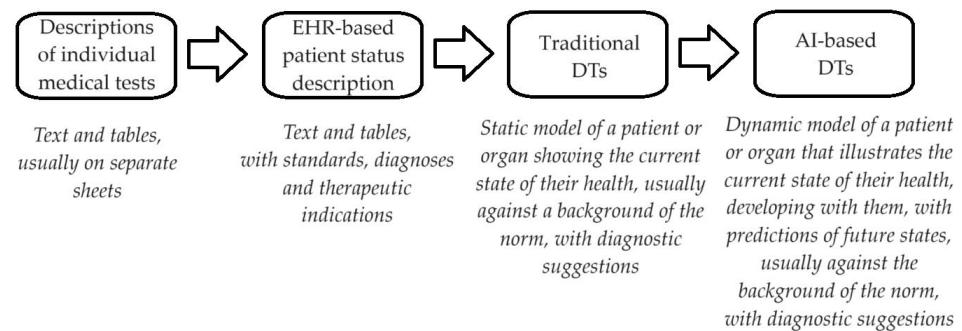


Figure 2. Evolution of AI-based DTs in healthcare [12–15].

2. Materials and Methods

2.1. Dataset and Devices

In developing the bibliometric analysis methods for this review, we focused on exploring the research landscape in the area of parameter selection strategies and development of AI-based DTs and possible innovations and clinical applications in this research area. For this purpose, we used bibliometric methods as analytical tools to review scientific publications. To better guide our review and selection of publications, we formulated research questions that help us discover key aspects of the research area, concerning research question 1 (RQ1)—the evolution of research topics/problems over time; RQ2—the geographical distribution of studies/publications, authors, scientific institutions, and publications with the highest impact; and RQ3—topics that may shape future research agendas. To achieve the above objectives, it was crucial to fully understand the current state of research, industry practices (connecting science and engineering, manufacturing and clinical applications), efforts, and future research directions in the pursuit of the optimal selection of AI-based DTs for a specific clinical application. Here, the analysis and interpretation of bibliometric data can make a key contribution to the ongoing discussion and build a more solid foundation for further analysis and research (Figure 3).

The criterion for selecting articles for review was also that we mainly wanted to use open sources in order to select not only new but also studies that are easily accessible for people from different fields, showing a broader perspective on AI-based DT. It is worth noting that not all publications were in English, and results published in national languages in India, China, South Korea, and Arab countries may not be available in other languages.

2.2. Methods

The study followed some of the points in the PRISMA 2020 guidelines [16] for creating a review with bibliographic analysis, namely point #3—rationale; point #4—objectives; point #5—eligibility criteria; point #6—sources of information; point #7—search strategy; point #8—selection process; point #9—data collection process; point #13a—synthesis methods; point #20b—synthesis results; and point #23a—discussion. In the review, we used the tools built into the Web of Science (WoS), Scopus, and PubMed databases, as well as the Biblioshiny tool from the Bibliometrics Rv.4.1.3 package for bibliometric analysis. This approach is better suited for bibliometric and scientometric studies, sometimes offering more precise categorization into conceptual/area/branch structures, authors, documents, and sources, and the different results are presented using graphs and information tables with a choice of analysis and visualization. Due to the complexity and interdisciplinary nature of the topic, we summarized the results of the review in a table.

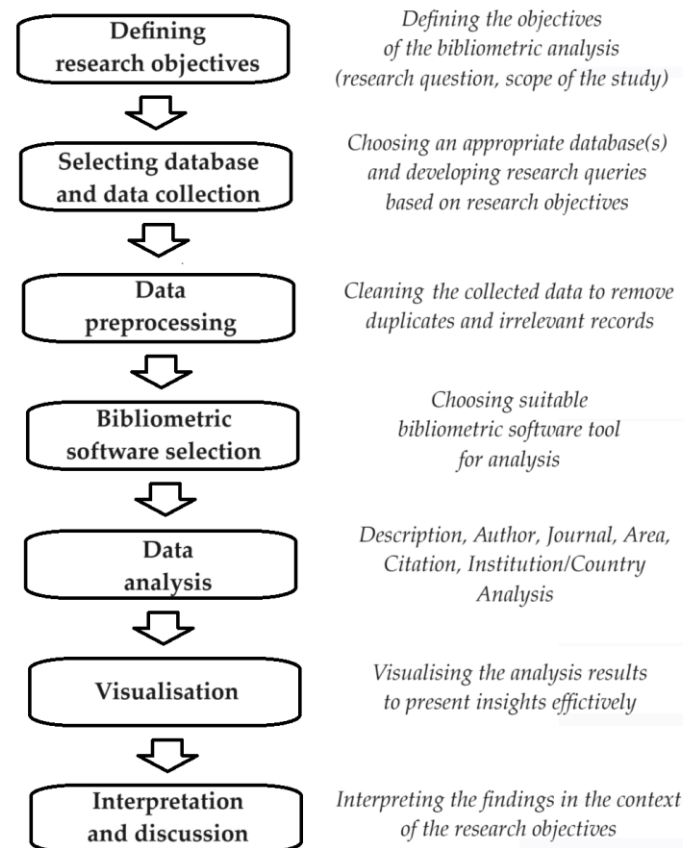


Figure 3. Bibliometric analysis procedure.

3. Results

After analysing the content of the main databases in the field of AI-based DT, we searched for research articles in three main databases: WoS, Scopus, and PubMed. This choice was dictated by the wide scope of research results they contained and the detailed data and analytical tools that allow for comprehensive analyses, which is particularly beneficial when conducting interdisciplinary bibliometric analyses and assessing the impact of research results, as was the case in our study. When creating advanced queries tailored to our research goals, we applied filters to select relevant publications. In this way, we limited our search to articles in English. Later in the study, we manually reviewed the articles again, excluding some of them (including duplicates) to match our research goals, which resulted in the final sample size (Figure 4). The WoS search was performed using “Topic” (the search was carried out using the set: title, abstract, keyword plus, and other keywords); the Scopus search was performed using the following set: article title, abstract, and keywords; and the PubMed search was performed using manual keyword sets.

Three databases were searched using the keywords “artificial intelligence” or “machine learning”, “digital twin” and “healthcare”. The obtained publications were subjected to further selection (Figure 4).

We started the bibliometric analysis with descriptive statistics to understand the characteristics of the dataset of the selected group of scientific publications, including leading authors, research institutions, topic areas, and emerging trends in the topic area under study (Figure 5). This allows us to identify evolving vocabulary and major research achievements. Examining changes in trends over time allows us to note changes in the focus/mainstream of research over time and the type and dynamics of the area, including the categorization of publications into topic clusters and a picture of the connections between research topics. This enabled the easier further identification of key topics and subdomains, including emerging research directions.

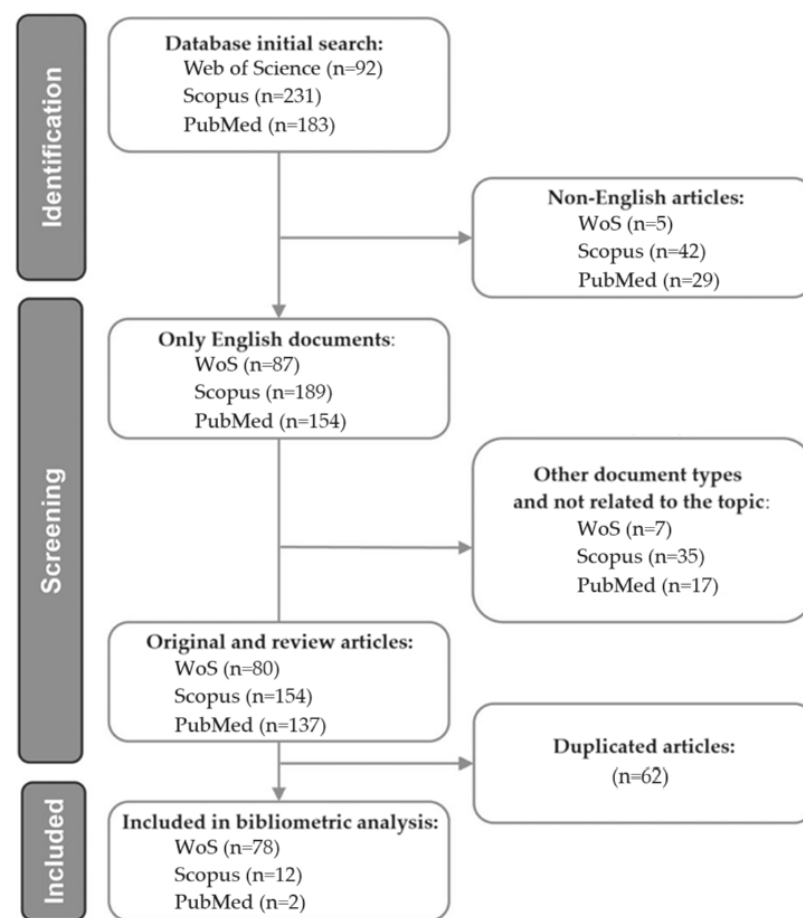


Figure 4. A PRISMA flow diagram of the review process using selected PRISMA 2020 guidelines [16]. Partial PRISMA 2020 checklist is added as Supplementary Materials.

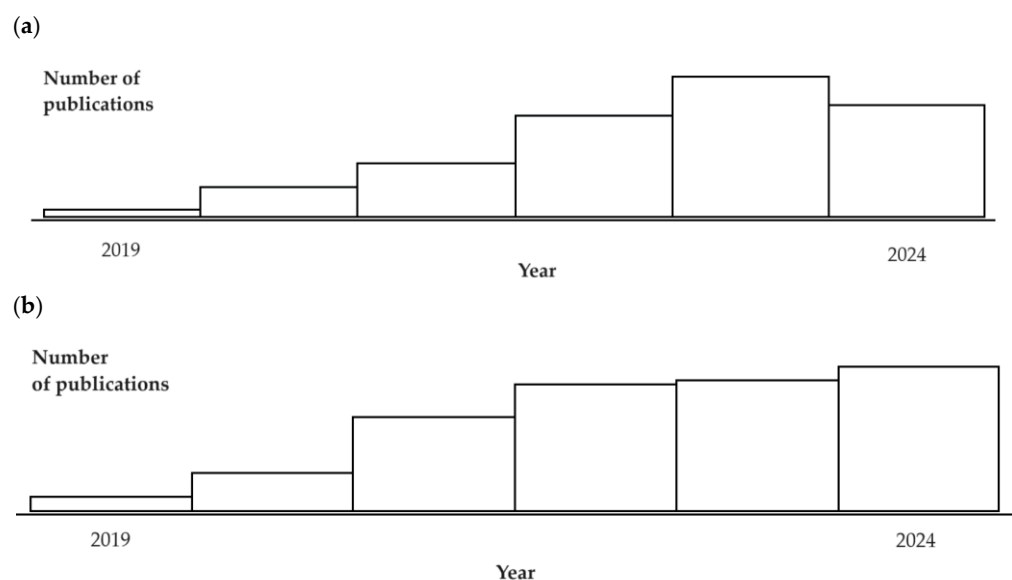


Figure 5. Review results: (a) AI + DT (183 publications, 2019–2024); (b) ML + DT (134 publications, 2019–2024).

The summary of the bibliographic analysis results is presented in Table 1. The largest number of research articles and almost three times fewer reviews were observed. A significant part of all publications were conference papers and book chapters. Most of the

publications were works in the field of computer science and engineering, with medicine only in third place. More than half of the leading topics were technological issues, and a smaller part was medicine. The table presents details of diseases and patient conditions that were addressed by the developed AI-based DTs. The leading countries conducting and financing research turned out to be large countries of Asia, North America, and Europe, which is also reflected in the leading affiliations and names of the leading researchers.

Table 1. Summary of results of bibliographic analysis (WoS, Scopus, PubMed).

Parameter/Feature	Value
Leading types of publication	Article (35.9%), conference paper (25.1%), book chapter (15.2%), review (13.9%)
Leading areas of science	Computer science (31.0%), Engineering (20.4%), Medicine (10.6%)
Leading topics	Industrial: design and manufacturing, telecommunications, security systems, human–computer interactions Medical: cardiac arrhythmia, neuromuscular disorders, nursing, tuberculosis and leprosy, wounds and ulcers, ECK, multiple sclerosis, high-level disinfection
Leading countries	India, USA, China, UK, Italy, Canada, Germany
Leading scientists	N. Wickramsinghe, A. Athar, J. Cai, T. Huynh-The, H.C. Kim, N. Ulapane, C. Yi, J. Zelcer
Leading affiliations	Swinburne University of Technology, Galgotias University, Mohamed Bun Zayed University of Artificial Intelligence
Leading funders (where information available)	National Natural Science Foundation of China, Horizon 2020 Framework Programme/European Commission, Ministry of Science and ICT of South Korea, Ministry of Science and Technology of China

3.1. Sources of Data for AI-Based DTs

The creation and maintenance of AI-based DTs in healthcare is facilitated by the availability of large historical datasets from previous clinical trials and other real-world data sources (e.g., patient biometrics collected from mobile devices). AI-based DTs in healthcare rely on a diverse set of data sources to accurately model a patient’s health status. Electronic health records (EHRs) provide a structured medical history, including diagnoses, treatments, and laboratory test results. Mobile devices and remote monitoring systems provide real-time physiological data such as heart rate, blood pressure, and activity levels. Medical imaging (e.g., MRI, CT scans) offers detailed anatomical and functional information that is crucial for creating accurate models of organs and systems. Genomic data provide personalized insights into a patient’s genetic predispositions and potential responses to treatment. Patient-reported outcomes and data from mobile health apps capture subjective experiences, symptoms, and lifestyle factors that impact health. Environmental and social determinants of health provide context about external factors, such as air quality and socioeconomic status, that affect a patient’s well-being. Pharmacy records provide information on medication adherence and potential interactions. Finally, clinical notes and unstructured data from healthcare professionals add valuable narrative information, often processed using natural language processing (NLP) techniques to help others understand a patient’s condition. There is also the challenge of combining disparate datasets and omics to quickly interpret large datasets to generate health and disease indicators to improve sampling and longitudinal analysis [17–20].

Multimomics in AI-based DTs involves integrating data from multiple “omics” disciplines—genomics, proteomics, transcriptomics, metabolomics, and epigenomics—to create a comprehensive and personalized model of a patient’s biological processes.

Genomics provides information about a patient's DNA sequence and genetic variants, helping to predict disease risk and response to treatment. Proteomics analyses the proteins expressed by genes, offering insight into cellular function and disease mechanisms. Transcriptomics examines RNA transcripts to understand patterns of gene expression that can indicate how genes are activated or repressed under specific conditions. Metabolomics focuses on metabolites, small molecules produced during metabolic processes, revealing real-time physiological states and treatment responses. Epigenomics examines DNA modifications that affect gene expression without changing sequence, providing information on how environmental factors affect genetic activity. By integrating these omics layers, AI-based DTs can model the complex interplay of biological processes with unprecedented precision. This holistic approach allows for the more accurate prediction of disease progression, treatment efficacy, and personalized healthcare strategies, ultimately improving patient outcomes [21–24].

3.2. AI-Based DT Architecture

An AI-powered patient DT is a virtual replica that dynamically reflects a patient's physiological, psychological, and behavioural states in real time. The architecture typically integrates data from multiple sources, such as EHRs, wearable devices, and imaging systems, to create a comprehensive patient model. The system uses advanced machine learning algorithms to continuously update and refine the twin, ensuring it accurately reflects the patient's current health status. A key component is a predictive analytics engine that can simulate potential outcomes based on different treatment scenarios, helping clinicians make informed decisions. The architecture often includes a feedback loop in which the AI analyses real-time data, predicts changes, and recommends interventions that are then validated and implemented in the real world. Interoperability is key, enabling the digital twin to integrate with various healthcare systems and devices, ensuring a seamless flow of data. Privacy and security measures, such as encryption and anonymization, are integral to protecting sensitive patient data. DT models also include a user interface that provides visualizations and insights for medical professionals, enabling them to more effectively monitor and manage patient health. In advanced implementations, the twin can incorporate natural language processing (NLP) to interpret unstructured data from clinical notes and patient communications, further enriching the model. Finally, continuous learning capabilities ensure that the DT model evolves over time, adapting to new medical knowledge and the changing health profile of an individual patient [22–27].

DTs represent a virtual model of a physical system, object, or process in real time, which is why their accuracy and functionality rely heavily on advanced AI models and ML methods and the accuracy of the data learned and used later for model honing. At the core of DTs are AI-based models such as deep learning, reinforcement learning, and generative models, which are used to predict and optimise the performance of physical/physiological entities. In healthcare applications, neural networks, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs), enable DTs to process sensor data, identify patterns, and predict future patient behaviour. These models are trained using large datasets that record historical performance, often relying on supervised learning methods to map inputs to specific outputs, such as predicting failures/illnesses or optimising operations. Unsupervised learning techniques such as clustering and anomaly detection are key in identifying patterns in unlabelled data, enabling DTs to detect unusual body system behaviour or emerging faults in real time. As a particularly effective method for decision-making in dynamic environments, reinforcement learning (RL) enables DTs to autonomously optimise processes such as autonomous systems, where the DT continuously improves its performance by interacting with its environment. Physics-based neural networks (PINNs) are also being used to embed physical laws directly into AI models, ensuring that predictions are consistent with known physical constraints, which is crucial in areas such as healthcare, aerospace, or energy systems, where deviations from physical laws can lead to dangerous consequences. The integration of generative models, such as

variational autoencoders (VAEs) and generative adversarial networks (GANs), enables the creation of virtual simulations that mimic the performance of a physical twin under different conditions, helping to proactively maintain or test new configurations without affecting actual operations. In addition, transfer learning allows DTs to quickly adapt to new environments or conditions, using knowledge from one model or system to optimise another. Real-time feedback loops between a physical entity and its digital counterpart are facilitated by AI models that process continuous streams of data, often involving complex sensor fusion techniques to combine data from different sources. A growing trend is the implementation of federated learning in DTs, where multiple twins collaborate to improve their models without sharing raw data, thus protecting privacy and reducing data transfer needs. DTs are evolving to use hybrid AI models that combine data-driven machine learning approaches with traditional physics-based simulations to provide more accurate, reliable, and interpretable results. These AI models allow digital twins to move from mere simulations to fully autonomous systems capable of self-adaptation, prediction, and decision-making in complex and changing environments.

There is no single formalized framework for AI-based DTs; moreover, depending on the application, the patient–organ model itself may not be the final result, but only an intermediate link for the further development of personalized medicines or implants (Figures 6–9) [28–32].

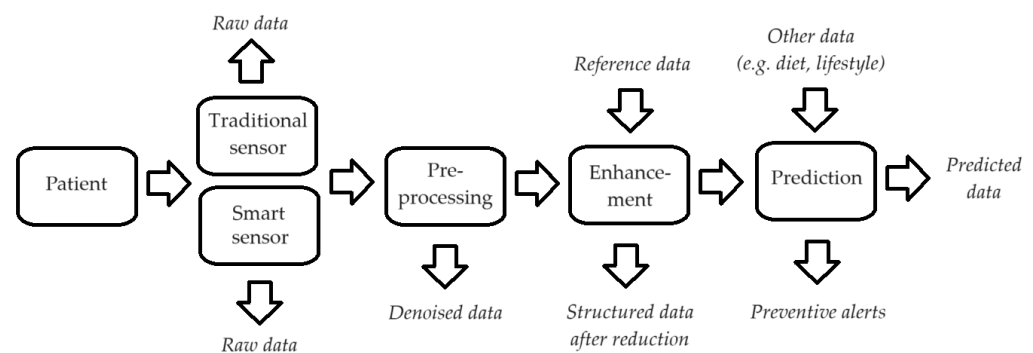


Figure 6. AI-based DT architecture (own version based on [22–27]).

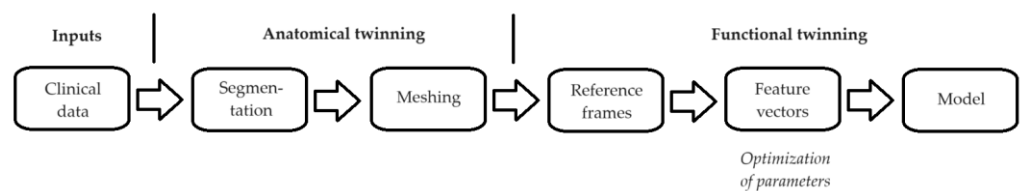


Figure 7. Example of basic workflow for AI-based DTs (own version based on [22–27]).

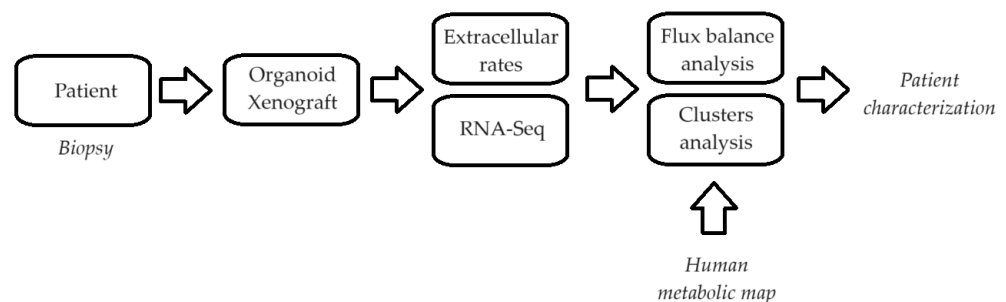


Figure 8. Example of advanced workflow for AI-based DTs (own version based on [22–27]).

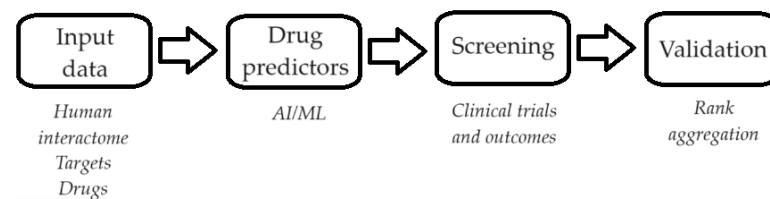


Figure 9. Example of pharmaceutical workflow for AI-based DTs for personalized drugs (own version based on [22–27]).

3.3. AI-Based DT Building

The proposed steps for building AI-generated DTs in personalized healthcare are as follows (Table 2). For the purposes of this study, it was assumed that DTs can be applied to both a sick patient and a healthy person (e.g., an athlete) in the context of preventive medicine [33,34]. Therefore, DTs must reflect both physiological and pathological processes/states.

Table 2. Stages of building AI-generated DTs in personalized healthcare (own version based on [33–39]).

Stage Number	Stage Name	Tasks
1	Data collection	Collecting comprehensive and diverse patient data, including electronic health records, genetic information, lifestyle data, and real-time data from wearable devices, to create a detailed foundation for the digital twin.
2	Data integration and preprocessing	The collected data undergoes integration and pre-processing, where data from different sources is standardized, cleaned, and organized (e.g., normalized, balanced) to ensure consistency and readiness for modelling.
3	Model/algorithm selection and development	AI algorithms and ML models are selected and developed to represent the patient's physiological systems, health status, and potential responses to treatment. This can be carried out based on pre-prepared initial templates for easier initialization.
4	Simulation and calibration	The DT is simulated using the processed data, and the model is calibrated to reflect the patient's actual current health status by adjusting parameters to ensure that the DT accurately reflects the patient's unique characteristics important for clinical diagnosis. Also, the directions and degree of their changes should be the same as those of the patient.
5	Validation and testing	The constructed DT undergoes rigorous validation and testing against real-world outcomes to ensure that it accurately predicts the patient's health trajectory and response to various interventions.
6	Personalization and adaptation	The DT is then personalized by incorporating patient-specific factors and continually updating the model with new data to reflect any changes in the patient's health over time and to keep the DT current (concurrent with the physical original).
7	Integration into clinical processes	The DT is integrated with clinical processes, providing healthcare providers with a tool for real-time decision support, treatment planning, and monitoring patient progress. Initially, this may be carried out using the infrastructure of medical simulation centres.
8	Feedback and refinement	As a DT is used in clinical practice, continuous feedback is collected from medical professionals and healthcare providers and results are monitored, leading to the continuous refinement and improvement of the model (both the technology itself, its engine and templates, and the specific DT of a healthy patient or one with specific conditions).
9	Ethical and regulatory compliance	Ensuring DT compliance with ethical standards and regulatory requirements is key, with attention paid to patient consent, data privacy, and legal issues.
10	Implementation and monitoring	Finally, the DT is implemented for use in personalized healthcare, with continuous monitoring to track its performance and effectiveness and to make further changes if necessary.
11	Continuous improvement	The DT is never finished; it constantly evolves with changes in technology and the patient's health, and after the patient dies and is anonymized, it serves as a basis for the teaching, learning, simulation, and development of the next generation of DTs.

The integrated digital twin model in personalised healthcare, generated by AI, is a dynamic, virtual representation of a patient's physiological and medical state. It is constantly evolving, reflecting real-time input from wearables, EHRs, and genetic information.

Artificial intelligence ensures model convergence by optimising algorithms that adapt and improve based on a variety of biomedical data, ensuring high quality and accurate simulations. Machine learning models dynamically adapt as new data are introduced, providing a personalised profile that becomes increasingly accurate over time. To systematically populate the model, artificial intelligence integrates structured (clinical data, lab results) and unstructured data (doctors' notes, imaging) using NLP and other AI techniques. The model predicts individual responses to treatment and medical conditions by simulating complex interactions in the patient's biological systems. Using patient-specific data helps doctors make more informed decisions about diagnosis, treatment plans, and disease prevention. The AI system continuously monitors convergence rates, ensuring that as the digital twin develops, its simulations are consistent with the patient's actual physiological responses. As more patients submit data, the AI learns from the larger dataset, further refining predictions and personalised care recommendations. Finally, the convergence of the model relies on a feedback loop between data collection, AI-based insights, and actual clinical outcomes, ensuring both the precision and accuracy of personalised healthcare.

The structure of such an integrated digital twin model in personalised healthcare consists of several interconnected layers: data acquisition, data processing, AI-based modelling, simulation, and feedback. Data acquisition includes the continuous collection of biomedical data from wearables, EHRs, and genetic databases, ensuring the capture of both real-time and historical data. The data processing layer cleans, organises, and standardises the incoming data, enabling consistent input for AI algorithms. AI-based modelling uses ML and deep learning techniques to interpret the data and generate personalised health models that adapt as more information becomes available. Layers of simulation then predict patient-specific outcomes, such as disease progression, response to treatment, or physiological changes.

3.4. AI-Based DT Clinical Use

DTs can use AI models to predict future health outcomes for individual patients in the form of an AI-generated digital twin to support the rapid assessment of intervention strategies *in silico*. We gain the ability to update real-time data from our physical patients and connect to multiple diagnostic and therapeutic devices. To date, at least several methods have been defined for the use of AI-generated DTs in personalized healthcare. By analysing a patient's DT, physicians can predict the onset of diseases or complications before they manifest in various scenarios, enabling early intervention and potentially preventing adverse health events and reducing the invasiveness and cost of therapeutic interventions. DTs can be used to simulate a patient's response to different treatment options, allowing healthcare providers to tailor treatments based on DT predictions, thereby optimizing outcomes for the individual. DTs can be used to model a patient's response to different medications, helping to identify the most effective medications and doses while minimizing side effects, especially in cases where personalized drug therapy is crucial. Surgeons can use DTs to plan complex surgeries by simulating different procedural approaches and predicting outcomes, thereby reducing risk and improving surgical precision. DTs enable the continuous monitoring of a patient's health through real-time data analysis, allowing for timely adjustments to treatment plans and the management of chronic conditions. Interestingly, patients can interact with their DTs to better understand their health status, the impact of lifestyle choices, and the potential outcomes of different treatments, allowing them to make more informed decisions.

With improved diagnostics and prognosis, AI has the potential to revolutionize various fields of medicine by supporting medical professionals for AI-cased DTs. The implementation of omni-channel engagement in digital health interventions and digital twins allows the flexibility of personalization that can increase and sustain patient engagement in digital health interventions and, ultimately, the quality of care and patient outcomes [40,41]. Yada et al. proposed to supplement structured cancer patient data with clinical concepts extracted from unstructured text (e.g., hospital discharge notes) using NLP techniques

to extract disease and drug names and link them to medical concepts using medical dictionaries. These data included richer information on patient symptoms than tabulated diagnosis records and could be a useful supplement, further enhanced by a drug safety module that captures suspected adverse drug reactions caused by the concomitant use of anticancer drug pairs [42]. In the study by Khan et al., DTs support the monitoring of respiratory failure, accelerating necessary interventions, helping to personalize treatment plans, and providing decision support to healthcare professionals. Respiratory data were collected using an ESP32 Wi-Fi Channel State Information (CSI) sensor and statistically augmented with time-series data to generate larger synthetic respiratory data with the comparative analysis of the experimental and synthetic datasets. These data were then denoised (smoothed and filtered) and dimensionality reduced using principal component analysis (PCA). The patient's breaths per minute (BPM) were estimated from the raw patient data with an accuracy of 92.3% from the raw respiratory data and an accuracy of 89.2% and 83.7% from the combined real and synthetic respiratory datasets with the larger synthetic dataset, respectively [43]. The study by Avanzato et al. demonstrates an approach to the diagnosis and treatment of chest diseases known as Lung-DT, which is based on IoT sensors and AI algorithms and is used to establish a digital representation of a patient's respiratory health. Based on the YOLOv8 neural network, this system classifies chest X-rays into five distinct lung disease categories (normal, COVID-19, lung opacity, pneumonia, and tuberculosis) with an average accuracy of 96.8%, a precision of 92%, a recall of 97%, and an F1 score of 94%. This not only allows for the real-time monitoring of lung health through continuous data collection from IoT sensors, facilitating early diagnosis and intervention, but also the automated and objective evaluation of chest X-rays, reducing the dependence on subjective human interpretation and combining multiple data streams for personalized treatment plans [44]. Maglogiannis et al. presented the AI4Work project approach to using digital twin technology to improve work environments where AI and robots seamlessly collaborate with humans in a human-machine task division across six domains. DTs can help occupational physicians and psychologists create a sustainable work environment, e.g., by monitoring the physical and mental health of hospital staff to predict burnout symptoms. In education, AI4Work can help maintain the mental health of teachers and students and a sustainable learning environment [45]. Kuriakose et al. showed that most of the research on DTs in healthcare has been carried out since 2018, with significant contributions from the interdisciplinary fields of AI, ML, and data science. It enables data interoperability and privacy protection and primarily represents an opportunity for chronic disease management, predictive analysis, drug discovery, and surgery planning [46]. Adibi et al. have shown that sensor-driven digital twins can further extend the capabilities of location-based services (LBS) in smart environments to meet both healthcare needs and patient requirements. The broader use of wearable medical devices, AI-based health analytics, and digital twin applications will help improve personalized healthcare interventions and emergency response capabilities. This demonstrates the potential of mobile DTs with patients [47]. In telemedicine, DTs can be used to deliver personalized care plans and real-time health monitoring to remote patients, ensuring that they receive high-quality care regardless of their location [46,47]. DTs enable virtual clinical trials, where multiple scenarios can be tested in a model before actual human trials, improving design and reducing risk in developing new therapies. By simulating multiple risk factors and their interactions, digital twins can provide personalized health risk assessments, helping to identify and mitigate potential threats to a patient's health. In post-operative or chronic care settings, DTs can be used to track a patient's rehabilitation progress, providing insight into the recovery process and adjusting care plans as needed to support optimal recovery. These benefits of AI-based DTs will see them dominate the healthcare market, displacing competing solutions [48–52].

It is possible to improve patient care through various means (simulated clinical trials, disease prediction, the remote monitoring of the patient's condition, treatment progress, treatment plan adjustments), especially in the environment of smart cities and smart territo-

ries and through the wider use of 6G, blockchain (and soon maybe quantum cryptography), and the Internet of Things (IoT), as well as in the field of medical technologies, such as multiomics. From a practical point of view, this requires not only efficient validation but also seamless integration with the existing healthcare infrastructure [53–58].

AI-based DTs in preventive medicine are used to proactively monitor and manage a patient's health, with the goal of preventing diseases before they occur. By continuously analysing real-time data from sources such as wearable devices and electronic health records, DTs can identify early warning signs and subtle changes in a patient's health. Predictive analytics within a twin model can predict the likelihood of developing specific conditions based on personalized risk factors such as genetics, lifestyle, and environmental influences. This includes early signs of aging or differences in fitness (but also risk) in athletes. These insights allow medical professionals to implement targeted interventions, such as lifestyle modifications or preventive treatments, to mitigate potential health risks. DTs can simulate the effects of different preventive strategies, helping clinicians choose the most effective approach for each person. It also supports personalized health education, providing patients with tailored recommendations to maintain optimal health. Overall, AI-based DTs enable a shift from reactive to proactive healthcare, significantly increasing efforts to prevent disease and maintain health [56–60].

3.5. User Interfaces

AI-based DTs in healthcare are characterized by user interfaces that are highly intuitive and visually engaging, designed to present complex data in an easily understandable format. They often include 3D representations of the human body, where clinicians can interact with different organs or systems, gaining real-time insight into the patient's condition. Interactive dashboards display a wealth of information, from vital signs to predictive analytics, using clear graphs, tables, and color-coded alerts to highlight critical issues. Interfaces are typically responsive, adapting to different devices such as tablets or desktops, ensuring accessibility in a variety of clinical environments. Advanced features can include simulation controls that allow healthcare providers to predict the outcomes of potential treatments by manipulating variables in the digital twin. Natural language processing capabilities are often integrated, allowing users to search the system using conversational language, making the technology more accessible to non-specialists. The interfaces also emphasize data privacy and security, with multi-layered authentication and encrypted data streams to protect patient information. Collaboration tools are embedded, allowing multiple healthcare providers to simultaneously view and interact with the DT, supporting a collaborative approach to patient care. Additionally, these interfaces are designed to be highly configurable, allowing clinicians to tailor the display to their specific needs, whether focusing on a specific organ system or integrating a specific patient history. They also offer real-time updates and notifications, informing users of any significant changes in a patient's condition or new insights generated by the AI algorithms. Finally, the interfaces often include educational modules, providing training and support to healthcare providers, ensuring they can effectively use the advanced features of the digital twin [61,62].

3.6. Interoperability

Interoperability between AI-based healthcare DT systems and other eHealth systems is key to seamless data exchange and integration. These systems are typically designed to communicate with EHRs, allowing DT systems to automatically retrieve patient data such as medical history, lab results, and imaging studies. Standard protocols such as HL7 and FHIR are often used to ensure consistent and accurate data exchange between the digital twin and various healthcare systems. AI-based DT systems are also capable of integrating with clinical decision support systems (CDSSs), enabling real-time insights to be incorporated into clinical workflows. Interoperability extends to mobile devices and remote monitoring tools, enabling the continuous transfer of patient data, which is then processed and visualized in the digital twin interface. These systems often include

application programming interfaces (APIs) that enable connections to other healthcare applications, facilitating the creation of a comprehensive and connected ecosystem. Data interoperability ensures that digital twins can be used across healthcare settings, from hospitals to outpatient clinics, ensuring continuity of care. They are also designed to meet regulatory standards such as HIPAA and GDPR, ensuring secure data exchange between systems and protecting patient privacy. Furthermore, AI-powered DTs can integrate with public health databases and research platforms, providing anonymized data for large-scale research and epidemiological tracking. Finally, interoperability with telehealth platforms enables the use of digital twins during remote consultations, providing clinicians with detailed patient information regardless of location [61–64].

4. Discussion

Previous studies and publications indicate that AI-generated DTs will play a key role in personalized healthcare by creating dynamic, real-time models of individual patients (or their organs), enabling precise and tailored flexible treatment plans. In a smart city environment, they will facilitate continuous monitoring and proactive health management, seamlessly integrating with digital health infrastructures such as IoT devices, telemedicine platforms, and entire eHealth systems. These AI-based models will enable the predictive diagnosis, early intervention, and optimization of healthcare resources, improving overall public health outcomes. In addition, DTs provide patients with personalized insights about their health, promoting informed decision-making and better engagement in their own care. In this way, DTs are a transformative tool in the transition towards more personalized, efficient, and proactive healthcare systems [63,64].

4.1. Current State of the Art Limitations

Healthcare DTs require diverse data from different sources, such as electronic health records, wearable devices, and genetic information. Integrating these different types of data into a coherent model remains a significant challenge. While AI can generate DTs, ensuring that they adapt (including through self-learning) in real time to reflect the changing condition of the patient remains difficult, especially in the case of unpredictable health events. Using sensitive personal data (including health data) to generate DTs raises privacy concerns. Ensuring that patient data are securely managed and compliant with regulations such as the EU GDPR is a critical limitation. The accuracy of AI-generated DTs depends on the quality and quantity of the data used. Inaccurate or incomplete data can lead to unreliable models, potentially resulting in incorrect diagnoses or health prognoses. Creating and maintaining complex DTs, especially those that simulate entire physiological systems, requires significant computing power that may not be readily available in all healthcare settings. In addition, there is no universal standard for how DTs should be built or used in healthcare, especially in the case of AI/ML, leading to inconsistencies in their development and application across organizations and systems. The use of DTs in healthcare raises ethical questions, such as who is liable if a DT makes an incorrect prediction and how to ensure equitable access to this technology. While DTs may work well at a small scale or for specific use cases, scaling these models to widespread use across patient populations is challenging due to the variability of individual health profiles. The complex algorithms behind AI-based digital twins can be difficult to interpret, making it difficult for healthcare providers to trust and act on the predictions and insights generated by these models. The regulatory framework for AI-based digital twins in healthcare is still evolving, and the approval processes, accountability, and long-term validation of these technologies are uncertain [65–67].

4.2. Directions for Further Research

Future research should focus on developing methods for the seamless (automatic) integration of different data sources, including standard formats and protocols, to improve the interoperability and consistency of DTs across healthcare systems. Exploring advanced

AI techniques for real-time data processing and dynamically updating DTs (reflecting the patient's current health status) will be critical to ensuring that these models accurately reflect the patient's current health status. Exploring ways to increase the accuracy of digital twins through improved data quality, more comprehensive datasets, and improved AI algorithms will be essential for reliable health predictions and interventions. Research should prioritize the development of robust privacy-preserving technologies such as federated learning, blockchain, and advanced encryption methods to protect sensitive patient data while enabling the generation of effective, reliable DTs. Exploring scalable, cost-effective computational methods, including the use of cloud computing and edge computing, will be critical to making DTs more accessible and feasible at all levels of healthcare. Research should delve into the ethical implications of digital twins, including establishing clear guidelines for accountability, transparency, and fairness in the use of AI-generated models in healthcare. Developing methodologies to scale digital twins to serve diverse and large patient populations, accounting for differences in genetics, lifestyle, and environmental factors, is a key area for future research. There is a need for research into explainable AI (XAI) approaches that can make digital twin models more understandable to healthcare providers, enabling them to trust and effectively use these tools in clinical decision-making. Conducting long-term studies to confirm the efficacy and safety of DTs in various clinical scenarios will be important for regulatory approval and widespread adoption. The ultimate goal is to explore how to effectively integrate DTs into existing clinical processes, including developing user-friendly interfaces and decision support systems that streamline rather than disrupt healthcare delivery [67–71].

The future of AI-based DTs is linked to emerging technologies such as ML, 6G, IoT, edge computing and augmented reality (AR), and the growing awareness of AI-based DTs among healthcare professionals and their patients. The development of legal medicine (health medicine) can have a major impact here. This future means even greater precision, speed, and accessibility of DTs, making it an integral part of eHealth and smart city and smart territory environments.

5. Conclusions

Digital society strategies in healthcare include the rapid development of digital twins (DTs) of patients and of human organs in medical research and clinical practice to more effectively, quickly, and cheaply use artificial intelligence (AI) to develop effective treatments. The evolution of digital twins from basic computer simulations to advanced, AI-based, real-time virtual representations has been driven by advances in computer science, data analysis, and data transmission and collection. Digital twins are now an essential tool in modern healthcare and are poised to play a key role in future clinical advances. Support for this form of personalized medicine is necessary due to the complex technological challenges, regulatory perspectives, and complex security and trust issues in this approach.

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